ECE 3710 – Quiz 4 – Autocorrelation and PSD

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Given the signal $g(t) = 4cos(2\pi 200t)$ and the following relationships:

$$R_g(\tau) = \frac{1}{T}\int_0^T g^*(t)g(t+\tau)dt \qquad S_g(\omega) = \mathscr{F}[R_g(\tau)]$$

$$\mathscr{F}[\cos(\omega_0 t)] = \pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)] \qquad \cos(a)\cos(b) = \frac{1}{2}[\cos(a-b)+\cos(a+b)]$$

1. (8 points) Find its autocorrelation function $R_g(\tau)$

$$\boxed{R_g(\tau) = 8\cos(400\pi\tau)}$$

2. (2 points) Find its Power Spectral Density $S_g(\omega)$.

$$\boxed{S_g(\omega) = 8\pi\,\delta(\omega-400\pi) + 8\pi\,\delta(\omega+400\pi)}$$

Problem 1 :

Step 1 : Autocorrelation

$$R_g(\tau) = \frac{1}{T}\int_0^T g^*(t)\, g(t+\tau)\,dt$$

* Remember $\underset{\text{A.C.}}{R(\tau)} \overset{\mathcal{F}}{\longleftrightarrow} \underset{\text{PSD}}{S_g(\omega)}$ *

$$R_g(\tau) = \frac{1}{T}\int_0^T \left[4\cos(2\pi 200 t)\cdot 4\cos(2\pi 200(t+\tau))\right] dt$$

* Remember, $\omega_0 = 2\pi\cdot 200 \rightarrow \omega_0 = 400\pi \rightarrow T = \frac{2\pi}{\omega_0} = \frac{1}{200}$

$$R_g(\tau) = \frac{1}{\frac{1}{200}}\int_0^{1/200} 16\cos(400\pi t)\cos(400\pi(t+\tau))\,dt$$

* Remember, $\cos(a)\cos(b) = \frac{1}{2}\left[\cos(a-b)+\cos(a+b)\right]$

$$R_g(\tau) = 200\int_0^{1/200} 16\cdot\frac{1}{2}\left[\cos\left(400\pi t - 400\pi(t+\tau)\right) + \cos\left(400\pi t + 400\pi(t+\tau)\right)\right]dt$$

$$R_g(\tau) = 1{,}600\int_0^{1/200} \underbrace{\left[\cos(-400\pi\tau)\right.}_{\text{Integrating this "constant" gives } t} + \underbrace{\cancelto{0}{\cos(800\pi t + 400\pi\tau)}}_{\text{Integrating this cos function tends to zero (area cancels)}}\left.\right]dt$$

$$R_g(\tau) = 1600\cos(-400\pi\tau)\cdot t\Big|_0^{1/200} = 1600\cdot\cos(-400\pi\tau)\cdot\frac{1}{200}$$

$$R_g(\tau) = 8\cos(-400\pi\tau) = 8\cos(400\pi\tau)$$

* cosine is even * $g(t) = g(-t)$

$$\boxed{R_g(\tau) = 8\cos(400\pi\tau)}$$

Step 2 : Find PSD

PSD $S_g(\omega)$ is the Fourier Transform Pair of Auto Correlation

$$S_g(\omega) = \mathcal{F}\left[8\cos(400\pi\tau)\right] = 8\left[\pi\,\delta(\omega-400\pi) + \pi\,\delta(\omega+400\pi)\right]$$

$$\boxed{S_g(\omega) = 8\pi\,\delta(\omega-400\pi) + 8\pi\,\delta(\omega+400\pi)}$$